Unit 4 · Session 2 · Thu, Nov 12, 2026

Endocrine Pharmacotherapy — Session 2

5 parts · every fact cited to its slide or textbook page. About 140 minutes of focused study.

Study plan

20 min
Read the guide
One pass, top to bottom. Bold = what the lecture emphasized.

70 min
Diagrams & tables
Redraw each flow from memory; cover a table column and recall it.

15 min
Cram sheet
Hide answers and quiz yourself. Repeat until it's automatic.

30 min
Self-test
Answer before you look. Read every explanation, even when right.

5 min
Objectives check
Tick each objective only if you can teach it out loud.

HOW TO READ THIS GUIDE

- **Bold** = emphasized on the lecture slides
- TB textbook only · SG course study guide only
- S12 = slide 12 · S12 (notes) = speaker notes · SG U1-R3 = study guide item · p. 17 = Katzung page
- Open **Sidekick** (bottom right) to search, listen, drill flashcards, quiz, or chat about this lecture

The whole lecture in 6 sentences

1. **Combined contraception** suppresses GnRH/FSH/LH; no estrogen with prior VTE/stroke, **smoker ≥ 35**, **migraine with aura**, estrogen-sensitive cancer, uncontrolled hypertension → progestin-only. Inducers (rifampin) need backup. S5; S6

2. **Tamoxifen** blocks breast but activates endometrium (report bleeding); **raloxifene** activates bone, blocks endometrium; both raise VTE. **Finasteride** lowers DHT and halves PSA. S7; S8; S9

3. **Insulin**: rapid (lispro, aspart), short (regular — the only IV insulin), NPH, basal (glargine, detemir, degludec); main risk **hypoglycemia** (rule of 15; beta-blockers mask it). S12; S13

4. **Metformin** first-line (avoid eGFR < 30, hold for contrast, B12); **sulfonylureas** highest oral hypoglycemia; **TZDs** fluid/heart failure; **SGLT2** genital infections, euglycemic DKA, cardiorenal benefit. S14; S15; S16

5. **GLP-1 RAs** (weight loss; no MTC/MEN2) and **DPP-4 inhibitors** (weight neutral) work glucose-dependently — low hypoglycemia; older adult who falls → DPP-4 or SGLT2. S18; S19; S21

6. **Hyperparathyroidism** → cinacalcet (watch hypocalcemia); **hypoparathyroidism** → calcitriol + calcium (watch hypercalcemia). S25; S27

Learning objectives

IO = syllabus unit objective · SO = this session's objectives slide.

IO4.1 Identify major pharmacologic classes used in the management of endocrine conditions and describe their mechanisms of action.

Part 1 · Gonadal Hormones & Inhibitors
Part 2 · Insulin
Part 3 · Non-Insulin Glucose-Lowering Drugs
Part 4 · Parathyroid Disorders

IO4.2 Describe the indications, contraindications, adverse effects, and clinically significant interactions associated with medications used to treat endocrine conditions.

Part 1 · Gonadal Hormones & Inhibitors
Part 2 · Insulin
Part 3 · Non-Insulin Glucose-Lowering Drugs

IO4.3 Compare pharmacologic agents within and across therapeutic classes to distinguish clinically relevant differences in efficacy, safety, and use.

Part 3 · Non-Insulin Glucose-Lowering Drugs
Part 5 · Putting It All Together: Endocrine Toolkit

IO4.4 Select appropriate pharmacologic therapy for common endocrine conditions based on clinical presentation and patient-specific factors.

Part 1 · Gonadal Hormones & Inhibitors
Part 3 · Non-Insulin Glucose-Lowering Drugs
Part 4 · Parathyroid Disorders
Part 5 · Putting It All Together: Endocrine Toolkit

IO4.5 Identify appropriate dosing, administration, and monitoring considerations for medications used in endocrine conditions.

Part 2 · Insulin
Part 3 · Non-Insulin Glucose-Lowering Drugs
Part 4 · Parathyroid Disorders

IO4.6 Explain important medication-use and safety information to patients receiving pharmacotherapy for endocrine conditions.

Part 1 · Gonadal Hormones & Inhibitors
Part 2 · Insulin
Part 5 · Putting It All Together: Endocrine Toolkit

SO1 Describe the mechanisms and clinical uses of gonadal hormone agents, including combined hormonal contraception, SERMs, and androgens/antiandrogens.

Part 1 · Gonadal Hormones & Inhibitors

SO2 Classify glucose-lowering drugs by mechanism and site of action, and compare agents on efficacy, hypoglycemia risk, and key adverse effects.

Part 2 · Insulin
Part 3 · Non-Insulin Glucose-Lowering Drugs

SO3 Distinguish hyperparathyroidism from hypoparathyroidism and identify the drug strategy used for each.

Part 4 · Parathyroid Disorders

SO4 Identify key contraindications, adverse effects, and drug interactions throughout.

Part 1 · Gonadal Hormones & Inhibitors
Part 3 · Non-Insulin Glucose-Lowering Drugs

SO5 Apply the full Endocrine pharmacology toolkit to integrated clinical scenarios, and complete a cumulative recall quiz spanning both sessions.

Part 5 · Putting It All Together: Endocrine Toolkit

ABOUT THIS LECTURE

- **PA PHARMACOLOGY COURSE · LECTURE 4 · SESSION 2 OF 2** — Endocrine Pharmacotherapy. **Session 2: Reproductive Hormones, Diabetes, and Bone Mineral Homeostasis | Thursday, November 12 | 8:30–11:00 AM** S1
- Recap: Session 1 (Nov 10) covered the hypothalamic-pituitary axis, thyroid/antithyroid drugs, and adrenocorticosteroid pharmacology. Session 1 gave us the axis. S1 (notes); S2; S2 (notes)
- Today the same negative-feedback logic explains contraception, then the many ways to lower glucose, finishing with the two opposite parathyroid problems — then integration and the cumulative recall quiz. S1 (notes); S2 (notes)
- Today picks up with gonadal hormones (Section 4) and runs through diabetes, parathyroid disorders, integration, and the cumulative recall quiz. S2 (notes)
- **Appendix 1 abbreviation key: PO = by mouth; SC = subcutaneous; IM = intramuscular; IV = intravenous; ER / XL = extended-release; BID = twice daily; mcg = microgram; U-100 etc. = units of insulin per mL; IUD = intrauterine device.** S36; S36 (notes)
- **DCIS = ductal carcinoma in situ; VTE / DVT / PE = venous thromboembolism / deep vein thrombosis / pulmonary embolism; MI = myocardial infarction; BPH = benign prostatic hyperplasia; PSA = prostate-specific antigen; PCOS = polycystic ovary syndrome.** S36; S36 (notes)
- **DKA = diabetic ketoacidosis; eGFR / CrCl = estimated glomerular filtration rate / creatinine clearance; CKD = chronic kidney disease; NPH = neutral protamine Hagedorn; SGLT2 = sodium-glucose cotransporter 2; DPP-4 = dipeptidyl peptidase-4; GLP-1 = glucagon-like peptide 1.** S36; S36 (notes)
- **MEN2 = multiple endocrine neoplasia type 2; CaSR = calcium-sensing receptor; PTH = parathyroid hormone; NYHA = New York Heart Association; CV = cardiovascular; CYP = cytochrome P450; QT = QT interval on ECG.** S36; S36 (notes)
- **APPENDIX 1** gives **Formulation, dose, indications, adverse effects, contraindications and toxicity** — your study reference: know the dose logic, the serious adverse effects, the contraindication that changes your decision, and the toxicity you would need to recognize. S36; S36 (notes); S37-66
- **Appendix 2: DRUG PROFILE · SUPPORTING** — supporting drugs explain interactions and rescue. **Listed for context; not profiled in detail.** S67; S67 (notes)
- **Original figures in this deck:** Figures 11, 15, 19, 21 and 22 (Combined Hormonal Contraception: Suppressing the HPG Axis; Pancreatic β-Cell Insulin Secretion: Secretagogue & Incretin Targets; Glucose-Lowering Drugs by Site of Action; S68
- HbA1c-Lowering Efficacy vs. Hypoglycemia Risk by Class; Parathyroid Disorders: Two Opposite Problems, Two Drug Strategies) were generated by the course director using the AI tool Claude

— original diagrams, not reproduced from Katzung's Basic and Clinical Pharmacology or any other copyrighted source. S68; S68 (notes)

- **Mechanism illustrations** (AI-generated, illustrative, not reproduced from copyrighted sources; schematic and not to scale): combined pills (Figure 12, slide 6); SERMs (Figure 13, slide 8); testosterone and 5-alpha-reductase inhibitors (Figure 14, slide 9); insulin signaling (Figure 16, slide 15); S68
- metformin (Figure 17, slide 17); SGLT2 inhibitors (Figure 18, slide 18); incretin drugs (Figure 20, slide 21); cinacalcet and the calcium-sensing receptor (Figure 23, slide 32). **Reference:** Anthropic. (2026). Claude (Sonnet 5) [Large language model]. https://claude.ai S68

SLIDE DISCREPANCIES

- **Rapid-acting insulin onset:** slides say about 15 minutes; the textbook gives 5–15 minutes. Same practical rule: dose with the meal. S12; p. 791

1 PART 1 · DRUG CLASS

Gonadal Hormones & Inhibitors

SO1 SO4 IO4.1 IO4.2 IO4.4 IO4.6

CLASS

- **Suppressing, replacing, or selectively modulating the reproductive axis.** Today the same negative-feedback logic explains contraception. S3; S3 (notes)
- **Figure 11. Original illustrative diagram (not from Katzung): exogenous estrogen and progestin suppress GnRH pulsatility and the FSH/LH surge, preventing the dominant-follicle development and LH surge required for ovulation.** S4; S4 (notes)
- This is the same negative-feedback template from Figure 1 (Session 1), now applied to the HPG axis. S4 (notes)

PROTOTYPE + OTHER DRUGS NAMED

- Combined oral contraceptives (ethinyl estradiol plus a progestin); progestin-only: norethindrone "minipill," depot medroxyprogesterone, levonorgestrel IUD; SERMs tamoxifen, raloxifene; testosterone; finasteride; spironolactone. S5; S7

MECHANISM OF ACTION

- **Mechanism: estrogen + progestin suppress GnRH/FSH/LH, preventing ovulation (Figure 11); progestin also thickens cervical mucus and thins the endometrium.** S5; S6 (notes)
- Chain (Figure 12): ethinyl estradiol plus progestin enter the bloodstream → hypothalamus lowers GnRH pulses; pituitary lowers FSH and LH → no follicle maturation and no LH surge, so no ovulation → result: pregnancy prevented (over 99% with perfect use). S6; S4 (notes)
- SERMs: tamoxifen or raloxifene bind the estrogen receptor → the receptor shape recruits different co-factors in each tissue. S8

INDICATIONS

- Combined contraception is available as oral pills, transdermal patch, or vaginal ring. S5
- Tamoxifen — antagonist in breast tissue (breast cancer treatment/prevention), agonist in endometrium (raises endometrial cancer risk). Tissue-selective activity is the defining feature of this class. S7; S7 (notes)
- Raloxifene — antagonist in breast/endometrium, agonist in bone (osteoporosis prevention) — ties back to Unit 2 bone pharmacology; prevents vertebral fracture; not used to treat breast cancer. S7; S8
- Testosterone — replacement for hypogonadism; monitor hematocrit (erythrocytosis risk) and lipids. Finasteride — 5-α-reductase inhibitor; reduces conversion of testosterone to DHT (BPH, androgenetic alopecia). S7

- Breast: blocked (antagonist), so cancer growth is slowed. Bone: activated (agonist), so bone density is kept. Endometrium: tamoxifen activates (cancer risk); raloxifene blocks. S8
- Testes make testosterone → 5-alpha-reductase converts it to DHT in prostate and skin → DHT drives prostate growth and hair loss → finasteride blocks 5-alpha-reductase → lower DHT: smaller prostate, less hair loss. S9; S9 (notes)

- Spironolactone — antiandrogen effect at high doses; used off-label for hirsutism/acne (same drug as the mineralocorticoid antagonist from Session 1): same molecule, different receptor, different clinical use. S7; S7 (notes); S9

ADVERSE EFFECTS (BOXED WARNINGS FIRST)

- Combined pills: risks include venous thromboembolism, hypertension and stroke. Common effects: nausea, breast tenderness, breakthrough bleeding. S6
- Tamoxifen raises endometrial cancer and VTE risk and causes hot flashes; patients should report vaginal bleeding. Raloxifene does not increase endometrial cancer, but raises VTE risk and hot flashes. S8; S8 (notes)
- Finasteride lowers PSA by about half, so double the PSA value when screening; sexual side effects occur. S9; S9 (notes)

CONTRAINDICATIONS

- **Contraindications: history of VTE/stroke, estrogen-sensitive cancer, uncontrolled hypertension, migraine with aura, and smokers age 35 or older (increased thromboembolic risk).** Liver disease too. S5; S5 (notes)
- Smoking at age 35+ substantially raises thromboembolic/cardiovascular risk with estrogen — the basis of the contraindication. S5; SG U4-R10
- "Smoker over 35" and "migraine with aura" are the classic exam triggers for choosing a progestin-only method — smokers 35 or older, migraine with aura and prior VTE rule out estrogen. S5 (notes); S6
- Progestin-only options (e.g., the "minipill," depot medroxyprogesterone (bone density warning), levonorgestrel IUD) avoid estrogen-related thromboembolic risk and are preferred when estrogen is contraindicated. S5; S6; S6 (notes)
- Both SERMs are contraindicated with prior VTE. Finasteride is teratogenic: pregnant women should not handle crushed or broken tablets. Testosterone: avoid in prostate cancer. S8; S9; S9 (notes)

DRUG INTERACTIONS

- Enzyme inducers (rifampin, carbamazepine, phenytoin, St. John's wort, some antiepileptics) lower contraceptive efficacy; use backup contraception. S5 (notes); S6; S67
- CYP2D6 inhibitors (paroxetine, fluoxetine) blunt tamoxifen activation. S8 (notes); S67

PHARMACOKINETICS

Not covered in materials

MONITORING

- Testosterone replacement: monitor hematocrit, PSA and lipids. S9; S9 (notes)

CLINICAL PEARL / WHY

- The contraindication list is the highest-yield content in contraception. S5 (notes)

Step by step

Can this patient take estrogen? S5; S6

Prior VTE or stroke? → Smoker aged 35 or older? → Migraine with aura? →

Estrogen-sensitive cancer, uncontrolled HTN, liver disease? →

Any yes → progestin-only (minipill, DMPA, levonorgestrel IUD)

Figure 11 · Combined hormonal contraception suppresses the HPG axis. Original illustrative diagram (not from Katzung): exogenous estrogen and progestin feed back on the hypothalamus and pituitary, suppressing GnRH pulses and the FSH/LH surge, so no dominant follicle and no ovulation. S4; S4 (notes)

Figure 12 · Combined pills: ovulation block and backup barriers. Schematic (illustrative) — Figure 12. Teal: hormone entry; blue: brain and ovary signaling; grey-green: local cervical and endometrial effects; red: outcome. Arrows show negative feedback: no ovulation, thick mucus, thin endometrium. S6; S6 (notes)

Figure 13 · SERMs: one receptor, different tissue effects. Schematic (illustrative) — Figure 13. Teal: receptor binding; blue and grey-green: tissue responses; green check = agonist, red X = antagonist; red outcome = risk in endometrium. Tamoxifen activates the endometrium (cancer risk); raloxifene blocks it. S8; S8 (notes)

Figure 14 · Testosterone, DHT and 5-alpha-reductase inhibitors. Schematic (illustrative) — Figure 14. Teal: start; blue: enzyme and hormone steps; green-grey: blocking drug; red: outcome, lower DHT. Finasteride blocks 5-alpha-reductase: smaller prostate, less hair loss, PSA roughly halved. S9; S9 (notes)

Contraceptive profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Combined oral contraceptives (ethinyl estradiol plus a progestin)	Estrogen plus progestin	Tablets (21/28-day packs); patch; vaginal ring	Ethinyl estradiol 20-35 mcg plus a progestin PO daily; patch weekly for 3 weeks; ring 3 weeks in, 1 week out	Contraception, cycle control, dysmenorrhea, acne
Norethindrone (progestin-only pill)	Progestin	Tablet 0.35 mg	0.35 mg PO daily at the same time, with no pill-free days	Contraception when estrogen is contraindicated, breastfeeding
Medroxyprogesterone depot (Depo-Provera)	Injectable progestin	IM 150 mg/mL; SC 104 mg/0.65 mL	150 mg IM (or 104 mg SC) every 12-13 weeks	Contraception, endometriosis pain, endometrial protection
Levonorgestrel IUD (Mirena)	Intrauterine progestin	Intrauterine system 52 mg (about 20 mcg/day)	One system placed in the uterus; effective up to 8 years for contraception	Contraception, heavy menstrual bleeding, endometrial protection

How to tell them apart. Sources: S37; S38; S39; S40

Contraceptive profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Combined oral contraceptives (ethinyl estradiol plus a progestin)	Nausea, breast tenderness, breakthrough bleeding, headache	VTE or stroke history, smoker 35 or older, migraine with aura, estrogen-sensitive cancer, uncontrolled hypertension, liver disease	VTE, stroke and MI risk; hyperkalemia with drospirenone; rifampin and some antiepileptics lower efficacy
Norethindrone (progestin-only pill)	Irregular bleeding, headache, acne, breast tenderness	Breast cancer, undiagnosed vaginal bleeding, liver tumor	Efficacy falls if a dose is more than 3 hours late; ectopic pregnancy risk if it fails
Medroxyprogesterone depot (Depo-Provera)	Irregular bleeding, weight gain, delayed return of fertility	Breast cancer, pregnancy, unexplained vaginal bleeding	Boxed warning: loss of bone mineral density; limit prolonged use; VTE
Levonorgestrel IUD (Mirena)	Irregular spotting, amenorrhea, ovarian cysts, pelvic pain	Pregnancy, pelvic infection, uterine anomaly, breast cancer	Uterine perforation, expulsion, infection; check strings

How to tell them apart. Sources: S37; S38; S39; S40

SERM and androgen profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Tamoxifen (Nolvadex)	Selective estrogen-receptor modulator	Tablet 10/20 mg	20 mg PO daily for 5-10 years in breast cancer	Breast cancer treatment and risk reduction, DCIS
Raloxifene (Evista)	Selective estrogen-receptor modulator	Tablet 60 mg	60 mg PO daily	Postmenopausal osteoporosis treatment and prevention, breast cancer risk reduction
Testosterone (cypionate, gel, patch)	Androgen	IM cypionate or enanthate 100-200 mg/mL; gel; patch; pellets	Cypionate 100-200 mg IM every 2 weeks (or 50-100 mg weekly); gel 50-100 mg daily	Male hypogonadism, delayed puberty
Finasteride (Proscar, Propecia)	5-alpha-reductase inhibitor	Tablet 5 mg (BPH); 1 mg (hair loss)	5 mg PO daily for BPH; 1 mg PO daily for male-pattern hair loss	BPH, androgenetic alopecia

How to tell them apart. Sources: S41; S42; S43; S44

SERM and androgen profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Tamoxifen (Nolvadex)	**Hot flashes, vaginal discharge, menstrual irregularity**	**History of DVT or PE, pregnancy, warfarin (for risk-reduction use)**	**Endometrial cancer, VTE, stroke, cataracts; CYP2D6 inhibitors (paroxetine, fluoxetine) reduce activation**
Raloxifene (Evista)	**Hot flashes, leg cramps, peripheral edema**	**VTE history, pregnancy**	**Boxed warning: VTE and fatal stroke risk; does not stimulate the endometrium**
Testosterone (cypionate, gel, patch)	**Acne, erythrocytosis, sleep apnea, gynecomastia, testicular atrophy**	**Prostate or breast cancer, polycythemia, pregnancy**	**Erythrocytosis, thrombosis, secondary exposure with gels; monitor hematocrit, PSA, lipids**
Finasteride (Proscar, Propecia)	**Decreased libido, erectile dysfunction, reduced ejaculate, gynecomastia**	**Pregnancy and women who may be pregnant (do not handle crushed tablets), hypersensitivity**	**Lowers PSA by about half; reports of depression; teratogenic to a male fetus**

How to tell them apart. Sources: S41; S42; S43; S44

WHY IT HAPPENS

- Why does a smoker over 35 get a progestin-only method? Estrogen adds clotting risk on top of smoking and age — the progestin-only options carry no estrogen. S5; S6

HIGH-YIELD

- No estrogen with prior VTE/stroke, smoker ≥ 35, migraine with aura, estrogen-sensitive cancer, uncontrolled hypertension → progestin-only. S5; S6
- Tamoxifen: endometrial agonist (report bleeding); raloxifene: bone agonist, endometrium blocked; both VTE. S7; S8

2 PART 2 · DRUG CLASS

Insulin

SO2 IO4.1 IO4.2 IO4.5 IO4.6

CLASS

- **Many mechanisms, one shared goal: glycemic control without excess hypoglycemia.** Today we cover the many ways to lower glucose. S10; S10 (notes)
- **Figure 15. Original illustrative diagram (not from Katzung): glucose entry raises ATP, closing K-ATP channels and triggering depolarization and calcium-dependent insulin release;** S11; S11 (notes)
- **sulfonylureas/meglitinides force channel closure directly, while GLP-1 receptor agonists and DPP-4 inhibitors amplify glucose-dependent release via the incretin pathway.** S11
- This is the mechanistic foundation for the insulin secretagogue and incretin-based drug classes that follow — the "glucose-dependent" nature of incretin-based therapy is exactly why these agents carry lower hypoglycemia risk than sulfonylureas. S11 (notes)
- TB Diabetes mellitus has four categories: type 1 (beta-cell destruction, absolute insulin deficiency), type 2, other specific types, and gestational diabetes (about 7% of US pregnancies). p. 789; p. 790
- TB Diagnosis: fasting plasma glucose ≥126 mg/dL or HbA1c ≥6.5%, confirmed by repeat testing. p. 790

PROTOTYPE + OTHER DRUGS NAMED

- Insulin lispro, aspart (rapid); regular (short); NPH (intermediate); glargine, detemir, degludec (long-acting). S12

MECHANISM OF ACTION

- **Mechanism: exogenous insulin activates the insulin receptor, promoting cellular glucose uptake and suppressing hepatic gluconeogenesis.** S12
- Chain (Figure 16): insulin binds the insulin receptor on muscle, fat and liver cells → receptor autophosphorylation starts the signal cascade → GLUT4 moves to the cell surface so glucose enters. S13; S13 (notes)
- Then the liver stores glycogen and stops making glucose → blood glucose falls (hypoglycemia is the main risk). S13

INDICATIONS

- Rapid-acting (lispro, aspart) — mealtime dosing; onset ~15 minutes. Short-acting (regular) — onset ~30 minutes. S12
- Intermediate-acting (NPH) — cloudy suspension; more variable absorption. Long-acting (glargine, detemir, degludec) — basal coverage; relatively peakless profile. S12
- Rapid-acting is given with meals; regular 30 minutes before. NPH peaks at about 4–10 hours; glargine, detemir and degludec are basal. Only regular insulin is given IV. S12 (notes); S13
- Sort any named insulin into rapid, short, intermediate or long-acting before discussing regimens — the scaffold for every insulin question. S12 (notes)
- TB Glucagon (IM, SC, IV, or 3-mg nasal powder) treats severe hypoglycemia when oral glucose is impossible; place the patient on their side (vomiting). p. 789

ADVERSE EFFECTS (BOXED WARNINGS FIRST)

- **Primary adverse effect across all insulins: hypoglycemia — counsel patients on recognition and treatment (rapid-acting carbohydrate).** S12
- Adverse effects: hypoglycemia, weight gain, lipohypertrophy, hypokalemia. Never share pens. S13
- TB Rotate injection sites to avoid lipohypertrophy; insulin allergy (IgE) is rare. p. 794; p. 795

CONTRAINDICATIONS

Not covered in materials

DRUG INTERACTIONS

- Beta-blockers can mask hypoglycemia warning symptoms. Glucagon and dextrose: rescue for severe hypoglycemia. S12 (notes); S13; S67

PHARMACOKINETICS

- TB Insulin preparations: lispro/aspart onset 5–15 min, 3–4 h duration; regular 30–60 min, 6–8 h; NPH 2–4 h, peak 6–7 h; glargine about 24 h, peakless. p. 791; p. 793

- **TB** Glargine is acidic (pH 4.0) — never mix it in the same syringe with other insulins. p. 793; p. 794

MONITORING

- Rule of 15 for hypoglycemia: 15 g carbohydrate, recheck in 15 minutes. S12 (notes); S13

CLINICAL PEARL / WHY

Not covered in materials

Figure 15 · β-cell insulin secretion: secretagogue and incretin targets. Original illustrative diagram (not from Katzung): glucose → ATP → K-ATP channels close → depolarization → Ca^{2+} → insulin release. Sulfonylureas/meglitinides force closure directly (any glucose); GLP-1 agonists and DPP-4 inhibitors amplify release only when glucose is high. S11; S11 (notes)

Figure 16 · Insulin: receptor signaling and time-action profiles. Schematic (illustrative) — Figure 16. Teal: insulin binding; blue: intracellular signaling and glucose uptake; grey-green: liver; red: outcome. Right panel shows onset, peak and duration by insulin type: rapid (lispro, aspart), short (regular), NPH (peak ~4–10 h), basal (glargine, detemir, degludec). S13; S13 (notes)

Insulin by time-action (S12)

Type	Examples	Onset / profile	Use
Rapid-acting	Lispro, aspart	~15 min	Mealtime
Short-acting	Regular	~30 min	30 min before meals; only IV insulin
Intermediate	NPH	Cloudy; peak ~4–10 h; variable	Basal/peak coverage
Long-acting	Glargine, detemir, degludec	Relatively peakless	Basal coverage

How to tell them apart. Sources: S12; S13

Insulin profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Insulin lispro (Humalog)	Rapid-acting insulin analog	SC or IV vial or pen 100 or 200 units/mL	Mealtime dose 0-15 minutes before eating; total daily insulin about 0.5-1 unit/kg; onset about 15 minutes	Type 1 and type 2 diabetes
Insulin aspart (NovoLog)	Rapid-acting insulin analog	SC or IV vial and pen cartridges 100 units/mL	Mealtime dose 0-15 minutes before eating; part of total daily insulin 0.5-1 unit/kg	Type 1 and type 2 diabetes, pump therapy
Regular insulin (Humulin R, Novolin R)	Short-acting human insulin	Vial 100 units/mL; U-500 concentrate; IV	SC 30 minutes before meals; IV infusion for DKA about 0.1 units/kg/hour	Diabetes mellitus, DKA and hyperkalemia (IV)
NPH insulin (Humulin N, Novolin N)	Intermediate-acting human insulin	Vial or pen 100 units/mL (cloudy suspension)	SC once or twice daily; basal start about 0.1-0.2 units/kg; peaks at 4-10 hours	Type 1 and type 2 diabetes, steroid-induced hyperglycemia
Insulin glargine (Lantus, Basaglar, Toujeo)	Long-acting basal insulin analog	SC pen or vial 100 or 300 units/mL	Start 10 units SC daily (or 0.1-0.2 units/kg) and titrate to fasting glucose	Type 1 and type 2 diabetes (basal)
Insulin detemir (Levemir)	Long-acting basal insulin analog	SC pen or vial 100 units/mL	10 units (or 0.1-0.2 units/kg) once or twice daily, titrated to fasting glucose	Type 1 and type 2 diabetes (basal)
Insulin degludec (Tresiba)	Ultra-long-acting basal insulin analog	SC pen 100 or 200 units/mL	10 units (or 0.1-0.2 units/kg) once daily; dosing interval flexible if at least 8 hours apart	Type 1 and type 2 diabetes (basal)

How to tell them apart. Sources: S45; S46; S47; S48; S49; S50; S51

Insulin profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Insulin lispro (Humalog)	Hypoglycemia, weight gain, lipohypertrophy, hypokalemia	Hypoglycemia, hypersensitivity	Severe hypoglycemia, hypokalemia; dose errors with concentrated pens; rotate sites
Insulin aspart (NovoLog)	Hypoglycemia, weight gain, injection-site reactions	Hypoglycemia, hypersensitivity	Severe hypoglycemia; hypokalemia; do not mix in a pump with other insulins

Drug	Adverse effects	Contraindications	Toxicity / overdose
Regular insulin (Humulin R, Novolin R)	**Hypoglycemia, weight gain, hypokalemia (IV)**	**Hypoglycemia**	**Only regular insulin may be given IV; U-500 dosing errors; monitor potassium**
NPH insulin (Humulin N, Novolin N)	**Hypoglycemia (nocturnal), weight gain, variable absorption**	**Hypoglycemia**	**Nocturnal hypoglycemia at the peak; mix suspension gently before use**
Insulin glargine (Lantus, Basaglar, Toujeo)	**Hypoglycemia, weight gain, injection-site reactions**	**Hypoglycemia, hypersensitivity**	**Hypoglycemia; do not mix with other insulins; U-300 is not interchangeable unit for unit**
Insulin detemir (Levemir)	**Hypoglycemia, weight gain, injection-site reactions**	**Hypoglycemia**	**Hypoglycemia; duration depends on dose; availability varies, so verify**
Insulin degludec (Tresiba)	**Hypoglycemia, weight gain, injection-site reactions**	**Hypoglycemia**	**Lower nocturnal hypoglycemia than glargine; steady state in 3-4 days; check pen strength**

How to tell them apart. Sources: S45; S46; S47; S48; S49; S50; S51

MNEMONIC (MEMORY AID)

- Rule of 15: 15 g fast carbohydrate, recheck in 15 minutes. S13

HIGH-YIELD

- Insulin's main risk is hypoglycemia; beta-blockers mask its warning signs; only regular insulin goes IV. S12; S13

3 PART 3 · DRUG CLASS

Non-Insulin Glucose-Lowering Drugs

SO2 SO4 IO4.1 IO4.2 IO4.3 IO4.4 IO4.5

CLASS

- Four non-incretin oral cards: metformin, sulfonylureas/meglitinides, TZDs, SGLT2 inhibitors; then the incretin classes (GLP-1 RAs, DPP-4 inhibitors) and the site-of-action map. S14; S14 (notes)
- **Figure 19. Original illustrative diagram (not from Katzung): glucose-lowering drug classes act at distinct sites — pancreatic β-cell, liver, muscle/fat, kidney, gut, and the incretin axis — each converging on overall blood glucose control.** S17; S17 (notes)

PROTOTYPE + OTHER DRUGS NAMED

- Metformin; glipizide, glyburide (sulfonylureas); repaglinide (meglitinide); pioglitazone (TZD); empagliflozin (SGLT2); exenatide, liraglutide, semaglutide (GLP-1 RA); sitagliptin, linagliptin (DPP-4). S14; S18

MECHANISM OF ACTION

- Metformin decreases hepatic gluconeogenesis: enters hepatocytes through OCT1 → inhibits mitochondrial complex I and activates AMPK → gluconeogenesis falls → insulin sensitivity improves in muscle → fasting glucose falls without hypoglycemia. S14; S15; S15 (notes)
- Sulfonylureas / meglitinides force K-ATP channel closure — insulin secretagogues (glipizide, glyburide, repaglinide). TZDs (pioglitazone) increase peripheral insulin sensitivity. S14
- Match each drug class to its mechanism/site: metformin — ↓ hepatic gluconeogenesis; sulfonylureas — force K-ATP channel closure (insulin secretagogue); TZDs — ↑ peripheral insulin sensitivity; SGLT2 inhibitors — block renal glucose reabsorption. SG U4-M; SG U4-R11

INDICATIONS

- Metformin: first-line for type 2 diabetes; no weight gain and low hypoglycemia risk. S14; S15; S15 (notes)
- SGLT2 inhibitors: cardiorenal benefit — benefits in heart failure and chronic kidney disease, beyond glucose lowering. S14; S16; S16 (notes)
- **Figure 21. Original illustrative comparison (illustrative relative values, not from Katzung): insulin and sulfonylureas show the greatest HbA1c-lowering efficacy but also carry the highest hypoglycemia risk,** S20
- **while metformin, DPP-4 inhibitors, and SGLT2 inhibitors show a more favorable efficacy-to-hypoglycemia-risk balance.** S20

- GLP-1 receptor agonists — glucose-dependent insulin release, slowed gastric emptying; DPP-4 inhibitors — prevent endogenous GLP-1 breakdown (incretin pathway). Sulfonylureas carry the highest hypoglycemia risk (glucose-independent insulin release). SG U4-M; SG U4-R11
- SGLT2 inhibitors block renal glucose reabsorption: glucose is filtered at the glomerulus → SGLT2 in the proximal tubule normally reabsorbs about 90% → empagliflozin blocks SGLT2. S14; S16
- Glucose and water are lost in the urine (glucosuria) → glucose, weight and blood pressure fall; infection risk rises. S16; S16 (notes)
- GLP-1 receptor agonists: glucose-dependent insulin release, slow gastric emptying, promote satiety/weight loss. DPP-4 inhibitors prevent breakdown of endogenous GLP-1, prolonging its glucose-dependent effect. S18; S18 (notes)
- Incretin chain (Figure 20): food in the gut releases GLP-1 → GLP-1 acts on the beta cell: glucose-dependent insulin release, less glucagon → DPP-4 normally breaks GLP-1 down quickly. S19; S19 (notes)
- Sitagliptin blocks DPP-4; semaglutide and liraglutide mimic GLP-1 → lower glucose, slower gastric emptying and less appetite. S19
- Figure 21 is the quantified summary of the drug-class discussion — it answers "why is metformin still first-line": not because it is the most potent, but because of its favorable risk profile. S20 (notes)
- TB Semaglutide lowers HbA1c 1.5–1.8%; liraglutide 3 mg and semaglutide 2.4 mg are approved for weight loss without diabetes. p. 800
- TB SGLT2 inhibitors lower HbA1c 0.5–1%, cause 2–5 kg weight loss, and reduce heart-failure events (dapagliflozin hazard ratio 0.74). p. 801; p. 802

ADVERSE EFFECTS (BOXED WARNINGS FIRST)

- Sulfonylureas: highest hypoglycemia risk among oral agents. TZDs: fluid retention/heart failure, weight gain. S14
- Metformin: GI upset is common — start low and take with food; vitamin B12 deficiency with long-term use; monitor B12 levels. S15; S15 (notes)

CONTRAINDICATIONS

- Metformin: avoid in significant renal impairment — lactic acidosis risk; avoid when eGFR is below 30; hold for contrast dye with reduced kidney function (iodinated contrast). S14; S15; S15 (notes); S67
- GLP-1 RAs: contraindicated with personal or family history of medullary thyroid carcinoma or MEN2 (thyroid C-cell warning). S18; S18 (notes); S19
- SGLT2 inhibitors: avoid for glycemic effect when eGFR is low. S16

- SGLT2 inhibitors: genital yeast (mycotic) infections, volume depletion, euglycemic DKA — hold when ill or before surgery; rare Fournier gangrene. S14; S16; S16 (notes)
- GLP-1 RAs: nausea/GI upset, vomiting, pancreatitis risk and gallstones. DPP-4 inhibitors are generally weight-neutral, well tolerated, with low hypoglycemia risk. S18; S18 (notes); S19
- **TB** Sulfonylureas: hypoglycemia and weight gain; elderly patients are at particular risk. p. 796; p. 797
- **TB** Acarbose (an alpha-glucosidase inhibitor, a drug that affects absorption of glucose): treat hypoglycemia with glucose, not sucrose. p. 799
- **TB** Metformin: full dose at eGFR 45–60, cautious at 30–45, contraindicated below 30; avoid in decompensated liver failure; check renal function at least annually. p. 798
- **TB** Thiazolidinediones are contraindicated in NYHA class III and IV heart failure. p. 799

DRUG INTERACTIONS

- Do not combine a GLP-1 RA with a DPP-4 inhibitor. Gemfibrozil raises repaglinide levels (CYP2C8). S19; S67

PHARMACOKINETICS

- **TB** Sitagliptin: reduce dose in renal impairment (50 mg at eGFR 30–50, 25 mg below 30). Exenatide is not approved below eGFR 30. p. 800; p. 801

MONITORING

Not covered in materials

CLINICAL PEARL / WHY

- Case: a 72-year-old with type 2 diabetes on metformin monotherapy, HbA1c above goal, lives alone, prior fall. **Which class of add-on therapy best balances efficacy and safety for this patient, and why?** S21; S21 (notes)
- **Given the fall history and the risks of hypoglycemia-induced falls, a DPP-4 inhibitor or SGLT2 inhibitor (low intrinsic hypoglycemia risk) is generally preferable to a sulfonylurea in an older adult living alone —** S21
- **matching drug choice to patient-specific risk, not just HbA1c-lowering potency.** S21
- This vignette applies Figure 21 directly to a geriatric safety scenario: "most potent" and "most appropriate" are not always the same answer. S21 (notes)
- Drug names reveal the class: **-gliptin** DPP-4 inhibitor (sitagliptin, linagliptin; low hypoglycemia risk); **-glutide** GLP-1 receptor agonist (liraglutide, semaglutide; weight loss, GI upset). S28; S28 (notes)
- **-gliflozin** SGLT2 inhibitor (empagliflozin; genital infections, euglycemic DKA); **-glitazone** thiazolidinedione (pioglitazone; fluid retention, heart failure); **-glinide** meglitinide (repaglinide; mealtime insulin secretagogue). S28

- **gli- (prefix)** sulfonylurea (glipizide, glyburide; K-ATP closure, hypoglycemia); **-ifene** SERM (tamoxifen, raloxifene; tissue-selective estrogen action); **-steride** 5-alpha-reductase inhibitor (finasteride; lowers DHT); **-gestrel** progestin (levonorgestrel). S28
- **-calcet** calcimimetic (cinacalcet; activates the CaSR, lowers PTH). A stem is a memory aid, not a rule: where no shared stem exists, learn the class name instead. S28
- **TB** SGLT2 inhibitors: not recommended for glycemic control below eGFR 45, contraindicated below 30 (for glucose). p. 802

Step by step

Add-on after metformin S16; S18; S21

HbA1c above goal on metformin → Older, lives alone, falls? → avoid sulfonylurea → Heart failure or CKD → SGLT2 inhibitor →

Weight loss wanted → GLP-1 RA (no MTC/MEN2) →

Low-risk, weight neutral → DPP-4 inhibitor

Figure 17 · Metformin: hepatic glucose output and AMPK. Schematic (illustrative) — Figure 17. Teal: drug entry; blue: AMPK signaling and glucose output; grey-green: muscle; short arrow = reduced gluconeogenesis; red: outcome — fasting glucose falls without hypoglycemia. S15; S15 (notes)

Figure 18 · SGLT2 inhibitors: glucose lost in the urine. Schematic (illustrative) — Figure 18. Teal: filtration; blue: tubule transport; green-grey: drug block; red: outcome, glucose lost in urine — lower glucose, weight and BP, with genital infection and euglycemic DKA risks. S16; S16 (notes)

Figure 19 · Glucose-lowering drugs by site of action. Original illustrative diagram (not from Katzung): β-cell (sulfonylureas, meglitinides), liver (metformin), muscle/fat (TZDs), kidney (SGLT2 inhibitors), gut and incretin axis (GLP-1 RAs, DPP-4 inhibitors) — all converging on blood glucose. S17; S17 (notes)

Figure 20 · Incretin drugs: gut, pancreas and brain. Schematic (illustrative) — Figure 20. Teal: gut signal; blue: beta-cell and enzyme steps; green-grey: drugs; red: clinical outcome. GLP-1 RAs mimic GLP-1; DPP-4 inhibitors protect it — lower glucose, slower gastric emptying, less appetite. S19; S19 (notes)

Figure 21 · HbA1c lowering vs. hypoglycemia risk. Original illustrative comparison (illustrative relative values, not from Katzung): insulin and sulfonylureas lower HbA1c most but carry the highest hypoglycemia risk; metformin, DPP-4 inhibitors and SGLT2 inhibitors balance efficacy and hypoglycemia risk more favorably. S20; S20 (notes)

Glucose-lowering classes (S14; S18)

Class	Example	Mechanism	Key risk
Metformin	Metformin	↓ Hepatic gluconeogenesis	GI, B12; lactic acidosis — avoid eGFR < 30
Sulfonylureas / meglitinides	Glipizide, glyburide, repaglinide	Force K-ATP closure	Highest oral hypoglycemia risk
TZDs	Pioglitazone	↑ Peripheral insulin sensitivity	Fluid retention/heart failure, weight gain
SGLT2 inhibitors	Empagliflozin	Block renal glucose reabsorption	Genital mycotic infection, euglycemic DKA
GLP-1 RAs	Exenatide, liraglutide, semaglutide	Glucose-dependent insulin; slow emptying	GI, pancreatitis; MTC/MEN2
DPP-4 inhibitors	Sitagliptin, linagliptin	Protect endogenous GLP-1	Weight-neutral; low hypoglycemia

How to tell them apart. Sources: S14; S18

Oral glucose-lowering profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Metformin (Glucophage)	**Biguanide**	**Tablet 500/850/1,000 mg; ER tablet; oral solution**	**Start 500 mg PO daily or BID with meals, titrate to 1,000 mg BID (max 2,550 mg/day)**	**Type 2 diabetes, prediabetes, PCOS (off-label)**
Glipizide (Glucotrol)	**Sulfonylurea (second generation)**	**Tablet 5/10 mg; XL 2.5/5/10 mg**	**5 mg PO daily 30 minutes before breakfast, titrate (max 40 mg/day; XL 20 mg)**	**Type 2 diabetes**
Glyburide (Diabeta, Micronase)	**Sulfonylurea (second generation)**	**Tablet 1.25/2.5/5 mg; micronized 1.5/3/6 mg**	**2.5-5 mg PO daily with breakfast, titrate (max 20 mg/day)**	**Type 2 diabetes**
Repaglinide (Prandin)	**Meglitinide**	**Tablet 0.5/1/2 mg**	**0.5-2 mg PO before each meal (skip if the meal is skipped), max 16 mg/day**	**Type 2 diabetes (mealtime)**
Pioglitazone (Actos)	**Thiazolidinedione (PPAR-gamma agonist)**	**Tablet 15/30/45 mg**	**15-30 mg PO daily, max 45 mg**	**Type 2 diabetes**

Drug	Class	Formulation	Adult dose	Indications
Empagliflozin (Jardiance)	SGLT2 inhibitor	Tablet 10/25 mg	10 mg PO once daily in the morning, may increase to 25 mg	Type 2 diabetes with cardiorenal benefit, heart failure, CKD

How to tell them apart. Sources: S52; S53; S54; S55; S56; S57

Oral glucose-lowering profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Metformin (Glucophage)	Nausea, diarrhea, metallic taste, vitamin B12 deficiency	eGFR below 30, acute metabolic acidosis, hypersensitivity	Lactic acidosis (renal impairment, contrast, alcohol, sepsis); hold around iodinated contrast; monitor B12
Glipizide (Glucotrol)	Hypoglycemia, weight gain, nausea, rash	Type 1 diabetes, DKA, severe liver disease; caution in sulfonamide allergy	Prolonged hypoglycemia (older adults, renal impairment); alcohol and beta-blockers worsen or mask it
Glyburide (Diabeta, Micronase)	Hypoglycemia, weight gain, nausea	Type 1 diabetes, DKA, bosentan co-use	Active metabolites cause prolonged hypoglycemia; avoid in older adults and CKD
Repaglinide (Prandin)	Hypoglycemia, weight gain, upper respiratory infection	Type 1 diabetes, DKA, gemfibrozil co-use	Hypoglycemia; gemfibrozil markedly raises levels (CYP2C8); CYP3A4 interactions
Pioglitazone (Actos)	Edema, weight gain, fractures, upper respiratory infection	NYHA class III-IV heart failure (boxed), active bladder cancer, liver disease	Heart failure exacerbation, fractures in women, possible bladder cancer; monitor weight and edema
Empagliflozin (Jardiance)	Genital yeast infections, UTI, volume depletion	Dialysis, hypersensitivity; not indicated in type 1 diabetes	Euglycemic DKA (hold when ill or fasting), Fournier gangrene, hypotension

How to tell them apart. Sources: S52; S53; S54; S55; S56; S57

Incretin drug profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Exenatide (Byetta, Bydureon)	GLP-1 receptor agonist	SC pen 5/10 mcg (Byetta); 2 mg weekly (Bydureon)	Byetta 5 mcg BID within 60 minutes before meals, then 10 mcg BID; Bydureon 2 mg SC once weekly	Type 2 diabetes

Drug	Class	Formulation	Adult dose	Indications
Liraglutide (Victoza, Saxenda)	GLP-1 receptor agonist	SC pen 18 mg/3 mL	Victoza 0.6 mg daily for 1 week, then 1.2-1.8 mg daily; Saxenda up to 3 mg daily	Type 2 diabetes, CV risk reduction, chronic weight management (Saxenda)
Semaglutide (Ozempic, Wegovy, Rybelsus)	GLP-1 receptor agonist	SC pen weekly; oral tablet 3/7/14 mg	Ozempic 0.25 mg weekly for 4 weeks, then 0.5-2 mg weekly; Rybelsus 3 mg daily, then 7-14 mg	Type 2 diabetes, CV risk reduction, obesity (Wegovy)
Sitagliptin (Januvia)	DPP-4 inhibitor	Tablet 25/50/100 mg	100 mg PO daily; 50 mg if eGFR 30-45; 25 mg if below 30	Type 2 diabetes
Linagliptin (Tradjenta)	DPP-4 inhibitor	Tablet 5 mg	5 mg PO daily; no renal adjustment	Type 2 diabetes

How to tell them apart. Sources: S58; S59; S60; S61; S62

Incretin drug profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Exenatide (Byetta, Bydureon)	Nausea, vomiting, diarrhea, injection-site nodules	Medullary thyroid cancer or MEN2 (Bydureon), CrCl below 30, severe gastroparesis	Pancreatitis, thyroid C-cell tumor warning (Bydureon), renal injury
Liraglutide (Victoza, Saxenda)	Nausea, vomiting, diarrhea, constipation, headache	Personal or family medullary thyroid cancer, MEN2, pancreatitis history	Boxed thyroid C-cell tumors, pancreatitis, gallbladder disease, dehydration with kidney injury
Semaglutide (Ozempic, Wegovy, Rybelsus)	Nausea, vomiting, diarrhea, constipation, abdominal pain	Personal or family medullary thyroid cancer, MEN2, hypersensitivity	Boxed thyroid C-cell tumors, pancreatitis, retinopathy complications, gallbladder disease, aspiration risk with anesthesia
Sitagliptin (Januvia)	Upper respiratory infection, headache, nasopharyngitis	Hypersensitivity (anaphylaxis, angioedema)	Pancreatitis, severe joint pain; low hypoglycemia risk when used alone
Linagliptin (Tradjenta)	Nasopharyngitis, cough, diarrhea	Hypersensitivity	Pancreatitis, bullous pemphigoid, arthralgia; eliminated mainly through bile

How to tell them apart. Sources: S58; S59; S60; S61; S62

SLIDE DISCREPANCY

- **Metformin and eGFR**
- Slides: Avoid when eGFR is below 30. S15

- Textbook: Full use at 45–60, cautious at 30–45 (lower doses), contraindicated below 30. p. 798

WHY IT HAPPENS

- Why do DPP-4 inhibitors rarely cause hypoglycemia? The incretin effect only boosts insulin when glucose is already high; sulfonylureas push insulin out regardless. S11; S35

HIGH-YIELD

- Metformin first-line (no weight gain, low hypoglycemia); avoid eGFR < 30; hold for contrast; B12. S14; S15
- Older adult with falls → DPP-4 or SGLT2, not a sulfonylurea. S21

4 PART 4 · DRUG CLASS

Parathyroid Disorders

SO3 IO4.1 IO4.4 IO4.5

CLASS

- **A brief recap of bone pharmacology, then a new angle: PTH itself.** Unit 2 covered osteoporosis drug therapy in depth: bisphosphonates, denosumab, teriparatide, and raloxifene (a SERM, also discussed again this session). S22; S23
- That therapy targets bone remodeling directly, generally in patients with normal parathyroid function. **This section takes a different angle: what happens — and what drugs are used — when parathyroid hormone secretion itself is the primary problem.** S23; S23 (notes)
- **If you need a refresher on bisphosphonate dosing or denosumab safety, revisit the Unit 2, Session 2 materials — that content is not repeated here.** S23
- This section is intentionally NOT re-teaching osteoporosis pharmacotherapy — it is new content on parathyroid pathology, so don't expect bisphosphonate-level detail here. S23 (notes)
- **Figure 22. Original illustrative diagram (not from Katzung): hyperparathyroidism (excess PTH, high calcium) is treated with cinacalcet to lower PTH,** S24
- **while hypoparathyroidism (deficient PTH, low calcium) is treated with calcitriol and oral calcium, sometimes with recombinant PTH 1-84 in severe cases.** S24
- Figure 22 is the organizing concept for the section — two opposite pathologies, two opposite drug strategies, both targeting the same hormone axis. S24 (notes)

PROTOTYPE + OTHER DRUGS NAMED

- Cinacalcet (calcimimetic); calcitriol (active vitamin D) plus calcium; recombinant human PTH 1-84. S25

MECHANISM OF ACTION

- Cinacalcet — a calcimimetic; activates the calcium-sensing receptor on parathyroid cells, reducing PTH secretion. S25
- Chain (Figure 23): parathyroid cells sense calcium through the CaSR → low calcium keeps PTH release high → cinacalcet makes the receptor more sensitive to calcium. S26
- The cell reads calcium as high and lowers PTH → PTH and serum calcium fall. S26; S26 (notes)
- Calcitriol (active vitamin D) plus oral calcium — mainstay for hypoparathyroidism, since PTH deficiency impairs vitamin D activation; calcitriol bypasses the PTH activation step. S25; S25 (notes)

INDICATIONS

- Cinacalcet: used for secondary hyperparathyroidism (e.g., in chronic kidney disease on dialysis) and parathyroid carcinoma. S25; S26
- Recombinant human PTH 1-84 — reserved for patients not adequately controlled on calcium/calcitriol alone. S25

ADVERSE EFFECTS (BOXED WARNINGS FIRST)

- Cinacalcet adverse effects: nausea, hypocalcemia and QT prolongation, so monitor calcium. S26

CONTRAINDICATIONS

Not covered in materials

DRUG INTERACTIONS

- Cinacalcet is a CYP2D6 inhibitor. Thiazides raise calcium, loops lower it. Vitamin D (cholecalciferol) replaces or supports calcitriol in deficiency. S26; S67

PHARMACOKINETICS

Not covered in materials

MONITORING

- Hyperparathyroidism — cinacalcet (lowers PTH secretion); hypoparathyroidism — calcitriol (bypasses the PTH-dependent vitamin D activation step) plus oral calcium. SG U4-R13

- Opposite monitoring targets: watch for hypocalcemia with cinacalcet, hypercalcemia/hypercalciuria with calcitriol/PTH replacement — a symmetry worth drawing out. S25; S25 (notes); S26

CLINICAL PEARL / WHY

- Case: a patient with chronic kidney disease is found to have an elevated PTH level with low-normal serum calcium. A second, unrelated patient who underwent recent thyroidectomy develops perioral numbness and low serum calcium with a low PTH level. S27
- **How does drug therapy differ between these two patients?** S27
- **The CKD patient has secondary hyperparathyroidism and may benefit from cinacalcet to reduce PTH secretion.** S27
- **The post-thyroidectomy patient has surgical hypoparathyroidism (inadvertent parathyroid gland injury) and needs calcitriol plus calcium replacement — the opposite pharmacologic strategy for the opposite underlying problem.** S27; S27 (notes)
- Post-thyroidectomy hypoparathyroidism is a classic, testable scenario and a bridge back to thyroid surgery from Session 1 (antithyroid drugs can also precede surgical management). S27 (notes)

Step by step

Which parathyroid drug? S25; S27

Check PTH and calcium →

High PTH (CKD, dialysis) → cinacalcet; watch hypocalcemia →

Low PTH, low calcium (post-thyroidectomy) → calcitriol + calcium →

Refractory → recombinant PTH 1-84 → Watch hypercalcemia on replacement

Figure 22 · Parathyroid disorders: two opposite strategies. Original illustrative diagram (not from Katzung): hyperparathyroidism (excess PTH, high calcium) → cinacalcet lowers PTH; hypoparathyroidism (deficient PTH, low calcium) → calcitriol plus oral calcium, sometimes recombinant PTH 1-84. S24; S24 (notes)

Figure 23 · Cinacalcet: tricking the calcium-sensing receptor. Schematic (illustrative) — Figure 23. Teal: sensing; blue: PTH steps; green-grey: drug; red: outcome, PTH and calcium fall. Cinacalcet sensitizes the CaSR so the cell reads calcium as high. S26; S26 (notes)

Hyperparathyroidism vs. hypoparathyroidism therapy (S25)

	Hyperparathyroidism	Hypoparathyroidism
Drug	Cinacalcet (calcimimetic)	Calcitriol + oral calcium; PTH 1-84 if refractory
How	Activates CaSR → ↓ PTH	Bypasses the PTH-dependent vitamin D activation
Typical patient	CKD on dialysis; parathyroid carcinoma	Post-thyroidectomy
Monitor for	Hypocalcemia	Hypercalcemia / hypercalciuria

How to tell them apart. Sources: S25; S26; S27

Parathyroid and calcium profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Cinacalcet (Sensipar)	**Calcimimetic (CaSR activator)**	**Tablet 30/60/90 mg**	**30 mg PO daily with food, titrate every 2-4 weeks (up to 180 mg/day) by PTH and calcium**	**Secondary hyperparathyroidism on dialysis, parathyroid carcinoma, primary hyperparathyroidism when surgery is not possible**
Calcitriol (Rocaltrol)	**Active vitamin D (1,25-dihydroxy)**	**Capsule 0.25/0.5 mcg; solution 1 mcg/mL; IV**	**Hypoparathyroidism: 0.25-2 mcg PO daily, titrated to calcium**	**Hypoparathyroidism, CKD secondary hyperparathyroidism, hypocalcemia**
Recombinant PTH 1-84 (Natpara)	**Parathyroid hormone replacement**	**SC pen cartridges 25/50/75/100 mcg**	**50 mcg SC daily in the thigh, adjusted by 25 mcg up to 100 mcg**	**Hypoparathyroidism not controlled on calcium and vitamin D**
Calcium carbonate or citrate	**Calcium supplement**	**Carbonate 500-1,250 mg tablets; citrate tablets**	**1,000-2,000 mg elemental calcium daily in divided doses (500-600 mg per dose)**	**Hypocalcemia, hypoparathyroidism, osteoporosis**

How to tell them apart. Sources: S63; S64; S65; S66

Parathyroid and calcium profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Cinacalcet (Sensipar)	**Nausea, vomiting, diarrhea, paresthesias, hypocalcemia**	**Hypocalcemia at start**	**Severe hypocalcemia (QT prolongation, seizures); CYP2D6 inhibitor; monitor calcium and PTH**
Calcitriol (Rocaltrol)	**Hypercalcemia, hypercalciuria, nausea, constipation**	**Hypercalcemia, vitamin D toxicity**	**Hypercalcemia, kidney stones, calcification; monitor calcium, phosphate, urine calcium**

Drug	Adverse effects	Contraindications	Toxicity / overdose
Recombinant PTH 1-84 (Natpara)	**Hypercalcemia, hypocalcemia, paresthesias, headache**	**Hypersensitivity; caution with osteosarcoma risk factors**	**Boxed osteosarcoma warning; severe hypocalcemia if stopped abruptly; availability is restricted, so verify current status**
Calcium carbonate or citrate	**Constipation, bloating, hypercalcemia**	**Hypercalcemia, hypercalciuria, kidney stones**	**Hypercalcemia (especially with calcitriol); carbonate needs food and acid (citrate for PPI users); separate from levothyroxine**

How to tell them apart. Sources: S63; S64; S65; S66

WHY IT HAPPENS

- Why calcitriol, not plain vitamin D, after thyroidectomy? Activating vitamin D needs PTH; calcitriol is already active. S25; S35

HIGH-YIELD

- High PTH (CKD) → cinacalcet, watch hypocalcemia; low PTH (post-thyroidectomy) → calcitriol + calcium, watch hypercalcemia. S25; S27

5 PART 5 · TOPIC

Putting It All Together: Endocrine Toolkit

SO5 IO4.3 IO4.4 IO4.6

KEY DEFINITIONS

- **Applying the whole Endocrine pharmacology toolkit.** The shared principle is negative feedback suppression by exogenous hormone. For each patient, name a first-line approach AND the single biggest safety concern. S29; S29 (notes); S30 (notes)

HOW IT WORKS

- **1.** The hypothalamic-pituitary-target gland axis is a shared template across thyroid, adrenal, and gonadal pharmacology — exogenous hormone therapy suppresses the levels above it via negative feedback. S31
- **2.** Antithyroid drug selection (methimazole vs. PTU) and glucocorticoid tapering both hinge on understanding exactly where and how each drug intervenes in its pathway. S31
- **3.** Glucose-lowering drugs act at distinct sites (pancreas, liver, muscle/fat, kidney, gut, incretin axis) — drug selection balances efficacy against hypoglycemia and other class-specific risks. S31
- **4.** Hyperparathyroidism and hypoparathyroidism require opposite drug strategies (lowering vs. raising effective PTH activity) — a new angle distinct from, but related to, the osteoporosis pharmacotherapy of Unit 2. S31
- **5.** Endocrine pharmacology is deeply interconnected — spironolactone, dopamine agonists, and SERMs each reappear across more than one organ system in this unit alone. S31
- Connecting back: osteoporosis pharmacology (Unit 2) and drug-metabolism/interaction concepts (Unit 1) both reappear here — pharmacology is cumulative, not siloed by unit. S31 (notes)

DRUGS NAMED AS EXAMPLES

- **A cumulative self-test spanning both sessions — use this to review before exams.** The recall quiz is a standalone resource, separate from in-lecture quick checks. S32; S32 (notes); S35 (notes)
- **Recall Q1:** PTU blocks both new thyroid hormone synthesis and peripheral T4-to-T3 conversion, giving it a faster effect on circulating active hormone than methimazole — important in thyroid storm. S33; S35
- **Recall Q2:** a patient on chronic glucocorticoid therapy abruptly stops the medication and develops hypotension and weakness: HPA-axis suppression leaves endogenous cortisol production unable to resume quickly. S33; S35
- Chronic exogenous glucocorticoid exposure suppresses CRH/ACTH via negative feedback (Figure 9), reducing the adrenal cortex's stimulus; abrupt withdrawal does not give the axis time to recover, precipitating adrenal insufficiency. S35
- **Recall Q3:** smoking at 35 or older contraindicates estrogen-containing contraception (thromboembolic and cardiovascular risk). **Q4:** sulfonylureas force K-ATP closure regardless of glucose; incretins act only when glucose is high. S33; S34; S35
- **Recall Q4 distractor:** DPP-4 inhibitors do affect insulin secretion — they prolong GLP-1's glucose-dependent effect. S34; S35
- **Recall Q5:** a patient with hypoparathyroidism after thyroid surgery needs treatment — the mainstay is calcitriol plus oral calcium: because PTH is required to activate vitamin D, calcitriol (already-active vitamin D) bypasses that activation step. S34; S35; SG U4-R13

- **Recall Q6:** SGLT2 inhibitors block renal glucose reabsorption, increasing urinary glucose excretion and creating a favorable environment for genital fungal overgrowth — a mechanism-based adverse effect. S34; S35; SG U4-R12

CLINICAL

- **Patient A — Graves disease, in thyroid storm:** PTU (blocks synthesis and peripheral T4-to-T3 conversion); check liver toxicity. S30; S35
- **Patient B — chronic prednisone user, abruptly out of refills:** restart the glucocorticoid and taper; the risk is adrenal insufficiency from HPA suppression. S30; S35
- **Patient C — type 2 diabetes, older adult with fall history, needs add-on therapy:** DPP-4 or SGLT2 inhibitor; avoid sulfonylurea hypoglycemia. S30; S21
- **Patient D — post-thyroidectomy, perioral numbness and low calcium:** calcitriol plus calcium; monitor for hypercalcemia. S30; S27

TRAP

- Choosing the most potent glucose-lowering drug instead of the safest for an older adult who falls. S21 (notes)
- Integrated case — diabetes: a DPP-4 or SGLT2 inhibitor is preferable to a sulfonylurea given the fall history — sulfonylureas force glucose-independent insulin release, raising hypoglycemia-driven fall risk. SG U4-S
- Integrated case — thyroid storm: beta-blocker + PTU first, then iodine solution only after PTU (never before — iodine first would supply substrate for new hormone synthesis before synthesis is blocked, worsening the crisis). SG U4-S
- Corticosteroids also help in thyroid storm by blunting peripheral conversion and covering relative adrenal insufficiency. SG U4-S
- Shared concept: effective endocrine drug selection depends on knowing exactly where in a pathway (or which receptor/site) a drug acts — that position determines both its efficacy and its specific risks. SG U4-S

Which drug would you choose? (S30)

Patient	Approach	Biggest safety check
A · Graves disease, thyroid storm	PTU	Liver toxicity
B · Chronic prednisone, out of refills	Restart and taper the glucocorticoid	Adrenal insufficiency
C · Older adult with type 2 diabetes and falls	DPP-4 or SGLT2 inhibitor	Avoid sulfonylurea hypoglycemia
D · Post-thyroidectomy, low calcium	Calcitriol + calcium	Hypercalcemia

How to tell them apart. Sources: S30; S35

Supporting drugs at a glance (S67)

Drug(s)	Why it matters here
Rifampin, carbamazepine, phenytoin	**CYP inducers that lower contraceptive efficacy**
Paroxetine and fluoxetine	**CYP2D6 inhibitors that blunt tamoxifen activation**
Glucagon and dextrose	**Rescue for severe hypoglycemia**
Beta-blockers	**Can mask hypoglycemia symptoms**
Iodinated contrast	**Hold metformin in reduced kidney function**
Gemfibrozil	**Raises repaglinide levels (CYP2C8)**
Bisphosphonates and denosumab	**Osteoporosis agents covered in Unit 2**
Thiazide and loop diuretics	**Thiazides raise calcium, loops lower it**
Vitamin D (cholecalciferol)	**Replaces or supports calcitriol in deficiency**

How to tell them apart. Sources: S67; S67 (notes)

MNEMONIC (MEMORY AID)

- Every case, two answers: **the approach** and **the one thing to check**. S30 (notes)

HIGH-YIELD

- Five takeaways: shared axis template · know where each drug intervenes · glucose drugs by site · opposite parathyroid strategies · cross-system drugs. S31

Cram sheet

Every term, mechanism, drug, and number. Hide the answers and quiz yourself; tap a blurred answer to peek.

Prompt	Answer
Estrogen contraindications	VTE/stroke, smoker ≥ 35, migraine with aura, estrogen-sensitive cancer, uncontrolled HTN, liver disease S5
Contraception + rifampin	Backup contraception S6; S67
Tamoxifen	Breast antagonist; endometrial agonist; CYP2D6 inhibitors blunt it S7; S8
Raloxifene	Bone agonist; endometrium blocked; VTE; not for breast cancer treatment S7; S8
Finasteride	↓ DHT; PSA halved → double it; teratogenic tablets S9
Testosterone	Monitor hematocrit, PSA, lipids S7; S9
Insulin buckets	Rapid lispro/aspart · short regular · NPH · basal glargine/detemir/degludec S12
Only IV insulin	Regular S12 (notes)
Hypoglycemia	Rule of 15; beta-blockers mask it; glucagon/dextrose rescue S13; S67
Metformin	AMPK, ↓ gluconeogenesis; eGFR < 30 avoid; contrast hold; B12 S15
Sulfonylurea risk	Highest oral hypoglycemia (forced K-ATP closure) S14; S35
TZD risk	Fluid retention, heart failure S14
SGLT2 risks	Genital yeast, volume depletion, euglycemic DKA, Fournier S16
GLP-1 RA	Weight loss; pancreatitis; no MTC/MEN2; never with DPP-4 S18; S19
Cinacalcet	CaSR activator; hypocalcemia, QT; CYP2D6 inhibitor S26
Hypoparathyroidism	Calcitriol + calcium; watch hypercalcemia S25; S35
Diabetes diagnosis	FPG ≥ 126 or HbA1c ≥ 6.5%, repeated p. 790
Glargine	pH 4.0 — never mix p. 793
Metformin by eGFR	45–60 full · 30–45 caution · < 30 contraindicated p. 798
TZD contraindication	NYHA III–IV heart failure p. 799
Acarbose hypoglycemia	Treat with glucose, not sucrose p. 799

Self-test

First, the lecture's own 6 knowledge checks — exactly as written. Then original PANCE-style questions. Tap an answer to reveal the explanation.

From the lecture

Lecture check 1 · S33; S35

Why is PTU, rather than methimazole, preferred in the management of thyroid storm?

- **A** PTU is better absorbed orally
- **B** PTU additionally blocks peripheral T4-to-T3 conversion
- **C** PTU has no hepatotoxicity risk
- **D** PTU works faster at the level of iodide trapping

ANSWER: B

PTU blocks both new thyroid hormone synthesis and peripheral T4-to-T3 conversion, giving it a faster effect on circulating active hormone than methimazole — important in the acute setting of thyroid storm. S33; S35

Lecture check 2 · S33; S35

A patient on chronic glucocorticoid therapy abruptly stops the medication and develops hypotension and weakness. What is the underlying mechanism?

- **A** Direct adrenal gland destruction by the drug
- **B** HPA-axis suppression leaving endogenous cortisol production unable to resume quickly
- **C** An allergic reaction to the glucocorticoid
- **D** Excess aldosterone production

ANSWER: B

Chronic exogenous glucocorticoid exposure suppresses CRH/ACTH via negative feedback (Figure 9), reducing the adrenal cortex’s stimulus to produce cortisol; abrupt withdrawal does not give the axis time to recover, precipitating adrenal insufficiency. S33; S35

Lecture check 3 · S33; S35

Which contraceptive counseling point reflects a true contraindication to estrogen-containing combined hormonal contraception?

- **A** Age under 20
- **B** Smoking at age 35 or older
- **C** Mild seasonal allergies
- **D** A family history of type 2 diabetes

ANSWER: B

Smoking at age 35 or older is a well-established contraindication to estrogen-containing contraception due to substantially increased thromboembolic and cardiovascular risk. S33; S35

Lecture check 4 · S34; S35

Why do sulfonylureas carry a higher hypoglycemia risk than DPP-4 inhibitors?

- **A** Sulfonylureas force K-ATP channel closure and insulin release independent of glucose level, unlike the glucose-dependent incretin pathway
- **B** DPP-4 inhibitors have no effect on insulin secretion
- **C** Sulfonylureas work only in type 1 diabetes
- **D** There is no difference in hypoglycemia risk between these classes

ANSWER: A

Sulfonylureas bind the K-ATP channel directly and force closure regardless of glucose level (Figure 15), while incretin-based agents amplify insulin release only when glucose is already elevated — explaining the difference in hypoglycemia risk. S34; S35

Lecture check 5 · S34; S35

A patient with hypoparathyroidism after thyroid surgery needs pharmacologic treatment. Which combination is the mainstay of therapy?

- **A** Cinacalcet alone
- **B** Calcitriol plus oral calcium
- **C** Bisphosphonate plus denosumab
- **D** Levothyroxine plus methimazole

ANSWER: B

Because PTH is required to activate vitamin D, hypoparathyroidism impairs that activation step; calcitriol (already-active vitamin D) bypasses it, and is given together with oral calcium as first-line therapy. S34; S35

Lecture check 6 · S34; S35

A patient taking an SGLT2 inhibitor develops a genital yeast infection. What is the mechanistic explanation?

- **A** Direct immunosuppression
- **B** Increased urinary glucose excretion promoting fungal overgrowth
- **C** Decreased renal blood flow
- **D** Systemic candidemia from the drug itself

ANSWER: B

SGLT2 inhibitors block renal glucose reabsorption, increasing urinary glucose concentration and creating a favorable environment for genital fungal overgrowth — a mechanism-based, well-known adverse effect. S34; S35

Original questions

Q1 · Clinical vignette · Gonadal Hormones & Inhibitors

A woman on a combined pill starts rifampin for latent TB. What counseling is needed?

- **A** Use backup contraception
- **B** No change; rifampin has no effect
- **C** Double the pill each day
- **D** Stop the pill during rifampin
- **E** Switch rifampin to St. John's wort

ANSWER: A

Enzyme inducers (rifampin, some antiepileptics, St. John's wort) lower contraceptive efficacy (S6). S6; S67

- **B.** It lowers efficacy (S6).
- **C.** Not the advice (S6).
- **D.** Backup, not stopping (S6).
- **E.** Also an inducer (S6).

Q2 · Mechanism · Gonadal Hormones & Inhibitors

Which statement about tamoxifen and raloxifene is correct?

- **A** Tamoxifen weakens bone
- **B** Raloxifene treats breast cancer
- **C** Both are breast agonists
- **D** Only raloxifene raises VTE risk
- **E** Only tamoxifen activates the endometrium

ANSWER: E

Tamoxifen is an endometrial agonist (cancer risk); raloxifene blocks the endometrium (S7–S8). S7; S8

- **A.** Bone agonist (S8).
- **B.** Not used to treat it (S8).
- **C.** Both block breast (S8).
- **D.** Both do (S8).

Q3 · Clinical vignette · Gonadal Hormones & Inhibitors

A man on finasteride for BPH has a PSA of 2.0 ng/mL. How should it be interpreted?

- **A** PSA is meaningless on finasteride
- **B** Stop finasteride for a month first
- **C** Double it — about 4.0 — when screening
- **D** Take it at face value
- **E** Halve it to about 1.0

ANSWER: C

Finasteride lowers PSA by about half, so double the value (S9). S9

- **A.** Still usable, doubled (S9).
- **B.** Not the taught approach (S9).
- **D.** It is halved (S9).
- **E.** Doubled, not halved (S9).

Q4 · Clinical vignette · Insulin

A patient on insulin and propranolol has a glucose of 45 mg/dL but no tremor or palpitations. Why?

- **A** The meter is reading falsely low
- **B** Hypoglycemia never causes tremor
- **C** Propranolol raises glucose
- **D** Insulin is no longer working
- **E** Beta-blockade masks warning symptoms

ANSWER: E

Beta-blockers can mask hypoglycemia warning symptoms (S13; S67). S13; S67

- **A.** Not the taught answer (S13).
- **B.** Tremor is a warning sign masked here (S13).
- **C.** Not the reason (S13).
- **D.** Glucose is low (S13).

Q5 · Mechanism · Insulin

Which insulin is the only one given intravenously?

- **A** Insulin glargine
- **B** Insulin degludec
- **C** NPH insulin
- **D** Regular insulin
- **E** Insulin detemir

ANSWER: D

Only regular insulin is given IV (S12 notes). S12; S47

- **A.** Basal, SC (S12).
- **B.** Basal, SC (S12).
- **C.** Cloudy suspension, SC (S12).
- **E.** Basal, SC (S12).

Q6 · Clinical vignette · Insulin

A patient on insulin is shaky with glucose 58 mg/dL and can swallow. What first?

- **A** 1 mg glucagon IM before any carbs
- **B** 15 g fast carbs, recheck in 15 min
- **C** Wait one hour, then recheck
- **D** Give the next insulin dose early
- **E** Skip the next meal and rest

ANSWER: B

Rule of 15 (S13). S13

- **A.** For patients who cannot take oral carbs (S67).
- **C.** Treat now (S13).
- **D.** Worsens it (S13).
- **E.** Treat now (S13).

Q7 · Clinical vignette · Non-Insulin Glucose-Lowering Drugs

A patient on metformin with eGFR 40 is scheduled for a CT with iodinated contrast. What should happen?

- **A** Switch to glyburide permanently
- **B** Add an SGLT2 inhibitor the same day
- **C** Continue without change
- **D** Hold metformin around the contrast study
- **E** Double metformin to cover the stress

ANSWER: D

Hold for contrast dye with reduced kidney function — lactic acidosis risk (S15; S67). S15; S67

- **A.** Not the taught step (S15).
- **B.** Not the answer (S16).
- **C.** Hold (S15).
- **E.** Raises risk (S15).

Q8 · Clinical vignette · Non-Insulin Glucose-Lowering Drugs

A patient with a family history of medullary thyroid carcinoma asks about semaglutide for diabetes and weight. Best answer?

- **A** Avoid GLP-1 RAs (C-cell warning)
- **B** Use it with a DPP-4 inhibitor
- **C** Safe if taken with food
- **D** Safe; it only affects the gut
- **E** Only liraglutide is a concern

ANSWER: A

GLP-1 RAs are contraindicated with personal or family history of MTC or MEN2 (S19). S18; S19

- **B.** Never combine (S19).
- **C.** Not relevant (S19).
- **D.** C-cell warning (S19).
- **E.** Class warning (S18).

Q9 · Mechanism · Non-Insulin Glucose-Lowering Drugs

Which class most often causes fluid retention and can worsen heart failure?

- **A** GLP-1 agonists (liraglutide)
- **B** DPP-4 inhibitors (sitagliptin)
- **C** SGLT2 inhibitors (empagliflozin)
- **D** Thiazolidinediones (pioglitazone)
- **E** Biguanides (metformin)

ANSWER: D

TZDs: fluid retention/heart failure, weight gain (S14). S14; S56

- **A.** Weight loss (S18).
- **B.** Weight neutral (S18).
- **C.** Benefit heart failure (S16).
- **E.** No fluid retention (S15).

Q10 · Clinical vignette · Non-Insulin Glucose-Lowering Drugs

A patient with type 2 diabetes has NYHA class III heart failure. Which drug is contraindicated?

- **A** Metformin
- **B** Semaglutide
- **C** Pioglitazone
- **D** Empagliflozin
- **E** Sitagliptin

ANSWER: C

Thiazolidinediones cause fluid retention and are contraindicated in NYHA class III and IV heart failure (p. 799). p. 799

- **A.** Not the HF contraindication (p. 798).
- **B.** Not contraindicated here (p. 800).
- **D.** Reduces heart-failure events (p. 802).
- **E.** Not contraindicated here (p. 801).

Q11 · Mechanism · Parathyroid Disorders

What should be monitored after starting cinacalcet?

- **A** Glucose for hypoglycemia
- **B** Potassium for hyperkalemia
- **C** TSH for hypothyroidism
- **D** Calcium for hypercalcemia
- **E** Serum calcium for hypocalcemia

ANSWER: E

Cinacalcet lowers PTH and calcium; monitor for hypocalcemia (and QT) (S25–S26). S25; S26

- **A.** Not the taught target (S26).
- **B.** Not the taught target (S26).
- **C.** Not the taught target (S26).
- **D.** That is calcitriol (S25).

Objectives checklist

Tick only what you could teach out loud. Saved on this device.

- [ ] **IO4.1** Identify major pharmacologic classes used in the management of endocrine conditions and describe their mechanisms of action.
- [ ] **IO4.2** Describe the indications, contraindications, adverse effects, and clinically significant interactions associated with medications used to treat endocrine conditions.
- [ ] **IO4.3** Compare pharmacologic agents within and across therapeutic classes to distinguish clinically relevant differences in efficacy, safety, and use.
- [ ] **IO4.4** Select appropriate pharmacologic therapy for common endocrine conditions based on clinical presentation and patient-specific factors.
- [ ] **IO4.5** Identify appropriate dosing, administration, and monitoring considerations for medications used in endocrine conditions.
- [ ] **IO4.6** Explain important medication-use and safety information to patients receiving pharmacotherapy for endocrine conditions.
- [ ] **SO1** Describe the mechanisms and clinical uses of gonadal hormone agents, including combined hormonal contraception, SERMs, and androgens/antiandrogens.

- [ ] **SO2** Classify glucose-lowering drugs by mechanism and site of action, and compare agents on efficacy, hypoglycemia risk, and key adverse effects.
- [ ] **SO3** Distinguish hyperparathyroidism from hypoparathyroidism and identify the drug strategy used for each.
- [ ] **SO4** Identify key contraindications, adverse effects, and drug interactions throughout.
- [ ] **SO5** Apply the full Endocrine pharmacology toolkit to integrated clinical scenarios, and complete a cumulative recall quiz spanning both sessions.